

Cheap, Stable, and High-Order Machine-Learning-Augmented Linear Multistep Methods for Quasi-Geostrophic Flows

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Abstract. The quality of a time integrator is governed jointly by computational and storage cost, stability, and accuracy. Implicit–explicit linear multistep methods (IMEX-LMMs) are attractive for pseudo-spectral solvers because they require only one nonlinear evaluation per step. High-order Runge–Kutta (RK) methods provide greater accuracy and stability at a higher computational cost. We introduce machine-learning-augmented IMEX-LMMs that retain the cost of an LMM of order p with step ΔT , while approaching the accuracy and stability of an RK method of order $q > p$ with step $\Delta t = \Delta T/K$, where $K \gg 1$.

Local Taylor expansions define the required correction and its radius of convergence. Direct evaluation, however, requires expensive high-order nonlinear derivatives. We approximate these terms with a deliberately shallow network. A Vandermonde layer reconstructs temporal derivatives with a closed-form error. A single convolutional layer approximates spatial derivatives and provides a local Wiener-filter correction for that error. The local taps are conditioned on the Taylor microscale of the state.

The neural network is embedded in the LMM solver and trained in rollout with a von Neumann-type penalty. Its shallow structure yields a closed-form model error and a closed-form scheme amplification factor, which provide explicit stability estimates. We demonstrate the method for two-dimensional quasi-geostrophic turbulence by augmenting an AB2–CN2 scheme toward fourth-order accuracy. A posteriori results demonstrate transfer across physical regimes and spatial and temporal discretizations. Stable time steps are $\mathcal{O}(10\text{--}50)$ times larger than the bare scheme’s CFL limit, while accuracy improves by $\mathcal{O}(100)$ at near-LMM cost.